

Andy Pham

SENIOR SOFTWARE ENGINEER

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SUMMARY

Experienced **Senior Software Engineer** with expertise in **C#, .NET Core, ASP.NET MVC, and Angular**, specializing in designing scalable, secure, and high-performance applications. Skilled in building **RESTful APIs, GraphQL endpoints**, and optimizing databases with **SQL Server and PostgreSQL**. Proficient in leveraging **microservices architecture, SOLID principles**, and cloud platforms like **Azure and AWS** to deliver robust solutions.

Strong background in implementing **CI/CD pipelines** using **Docker, Kubernetes, and Azure DevOps**, as well as event-driven systems with **RabbitMQ and Apache Kafka**. Experienced in **TDD** and testing frameworks like **NUnit, xUnit, and Jest**, ensuring reliable, maintainable code. A collaborative team player, mentor, and Agile practitioner, adept at modernizing legacy systems and delivering business-focused solutions.

KEY SKILLS

- **Languages:** C#, C++, JavaScript (ES6+), Typescript, SQL, PHP
- **Frameworks/Libraries:** .NET Core, C#.NET, ASP.Net, Entity Framework, WEB API, Restful Services, Web API WCF, ADO.NET, VB.NET, Blazor, Node
- **Web Technologies:** Angular, HTML5, CSS3, jQuery, Bootstrap, Tailwind CSS, Bootstrap, Mobx, RxJS, Context API, Web Services, XML, XSL, XSLT, IIS, XAML and AJAX
- **Paradigm:** Microservice, SOLID principle, Singleton Pattern, Factory Pattern, OOP, MVC
- **Cloud Services:** Azure, AWS, Serverless Computing, Edge Computing
- **Database:** MS SQL Server, PostgreSQL, LINQ, CosmosDB
- **CI/CD:** Docker, Kubernetes, Jenkins, Azure DevOps
- **Message Brokers:** Rabbit MQ, Apache Kafka
- **Testing:** NUnit, xUnit, MS Test, Jest, Cypress, Test Driven Development
- **Documentation & API:** Swagger, MS Sharepoint, Confluence, SOAP, Restful API, GraphQL
- **Agile Methodologies:** Agile, Scrum

EDUCATION

Aug 2011 – May 2015 → **Bachelor's Degree in Computer Science**

University of Michigan | ANN Arbor, MI

GPA: 3.7/4.0

RELEVANT EXPERIENCE

Feb 2021 – Jan 2025 → Senior Software Engineer

Deloitte | Lansing, MI

- Led the modernization of a **legacy system for a global financial institution**, leveraging **Angular, MobX, RxJS, and TypeScript** for the frontend, while utilizing **.NET Core, C#, and ASP.NET MVC** for the backend to ensure high performance and scalability.
- Optimized backend processes using **C#**, leading to a 30% improvement in system performance.
- Designed and implemented **RESTful APIs with .NET Core and C#**, enabling seamless integration between the frontend and backend.
- Developed dynamic and interactive user interfaces by utilizing **Angular's** reactive programming capabilities and observables.
- Integrated third-party services through **Angular's HttpClient** and **ASP.NET Core's Dependency Injection**, ensuring robust API interactions.
- Created reusable **Angular components, services, and modules**, employing **Reactive Forms, Directives, and Custom Pipes** to enhance UI responsiveness.
- Architected RESTful Web APIs and **microservices in .NET Core**, streamlining data exchange and backend efficiency.
- Implemented **server-side Blazor** to enhance rendering performance and optimize backend operations.
- Applied **design patterns** such as Singleton, Factory, and Dependency Injection to refactor backend code, improving modularity and maintainability.
- Developed and optimized **T-SQL scripts, stored procedures, triggers, and views** for Microsoft SQL Server, ensuring data integrity and efficient querying.
- Migrated and deployed applications to **Azure Cloud**, utilizing Azure App Service, Azure SQL Database, and Azure Blob Storage for secure cloud storage and management.
- Engineered authentication and authorization mechanisms using **ASP.NET Identity and OAuth**, ensuring application security.
- Established a **CI/CD pipeline** using **Docker, Kubernetes, Jenkins, and Azure DevOps**, automating deployments and streamlining development workflows.
- Enforced strong typing and object-oriented principles with TypeScript, reducing runtime errors by **50%** and enhancing code maintainability.
- Managed the **entire Software Development Life Cycle (SDLC)** from requirements gathering to deployment, ensuring seamless delivery of software solutions.
- Utilized **Git** for version control, maintaining organized and structured repositories.
- Advocated for **Test-Driven Development (TDD)** and **Behavior-Driven Development (BDD)** to maintain high code quality and reduce defects.
- Fostered collaboration using **SharePoint and Confluence**, streamlining document management and knowledge sharing.
- Led the transition from **monolithic architectures to microservices**, improving system scalability, flexibility, and maintainability.
- Played an active role in **agile methodologies**, facilitating **Scrum and Kanban** practices such as sprint planning, stand-ups, and retrospectives.
- Provided **technical mentorship** to junior developers, fostering a culture of continuous learning and innovation.
- Partnered with cross-functional teams to define project scope, ensuring timely delivery of high-quality solutions.

Aug 2017 – Feb 2021 → Software Engineer

Infosys Consulting | Chicago, IL

- Led the development of an **e-commerce platform**, utilizing **.NET Core, C#, and ASP.NET MVC** for the backend and **Angular** for the frontend.
- Implemented a **microservices architecture**, enhancing system modularity and scalability.
- Integrated **server-side Blazor**, significantly improving rendering performance.
- Developed **reactive and asynchronous UI components** using **RxJS, Angular CLI, and state management tools like NgRx**.
- Designed efficient **routing strategies** in Angular, enabling seamless navigation between components.
- Applied **SOLID principles, MVVM, and object-oriented design** to build scalable, maintainable applications.
- Developed robust **backend services** using **.NET Core**, ensuring secure data processing and business logic execution.
- Leveraged **Entity Framework and LINQ** to enhance database interactions and connectivity.
- Improved application security with **ASP.NET Identity and OAuth**, ensuring secure user authentication and authorization.
- Utilized **AWS and Azure** to enhance deployment efficiency, implementing **CI/CD pipelines** for streamlined development processes.
- Conducted **performance optimizations**, significantly reducing application response times and enhancing user experience.
- Implemented **GitHub Actions-based CI/CD pipelines**, automating testing, integration, and deployment workflows.
- Integrated **Azure Event Grid, Service Bus, and Event Hub** to ensure **reliable event-driven processing**.
- Strengthened code quality through **Swagger, SharePoint, and Confluence documentation**.
- Utilized **Jira and Trello** for agile project management, tracking progress and ensuring timely delivery.

Sep 2015 – Aug 2017 → **Software Developer**
Humana | Troy, MI

- Played a key role in the development of a **healthcare portal**, employing **Angular for the frontend** and **.NET for the backend** to create an intuitive user experience.
- Built **responsive and accessible UIs** using **Angular, HTML, CSS3, and TypeScript**.
- Applied **object-oriented programming (OOP) concepts and design patterns** in C#, ensuring modular and maintainable code.
- Developed **RESTful APIs and microservices** using .NET Core and C#, enabling seamless backend-frontend communication.
- Integrated **real-time tracking, payment processing, and order management APIs**, enriching app functionality.
- Utilized **Scrapy** for social media data extraction, enhancing analytical insights.
- Contributed to **frontend development optimization**, adhering to JavaScript best practices.
- Gained experience with **headless CMS solutions**, integrating **DatoCMS with Node.js**.
- Used **Git for version control**, facilitating smooth collaboration and code management.
- Assisted in writing **unit tests with Jasmine/Karma** and backend tests with NUnit to ensure application reliability.
- Deployed applications to **Azure Cloud**, gaining hands-on experience with cloud infrastructure and deployment strategies.
- Worked within **Scrum and Kanban teams**, collaborating effectively to deliver features and enhancements.